

DEEP LEARNING INDABAX SOUTH AFRICA 2019

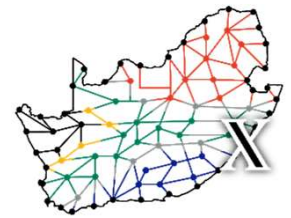
Report



14 - 17 APRIL 2019

UNITE Building
University of KwaZulu-Natal
Durban

DEEP LEARNING INDABA^X SOUTH AFRICA 2019



INTRODUCTION

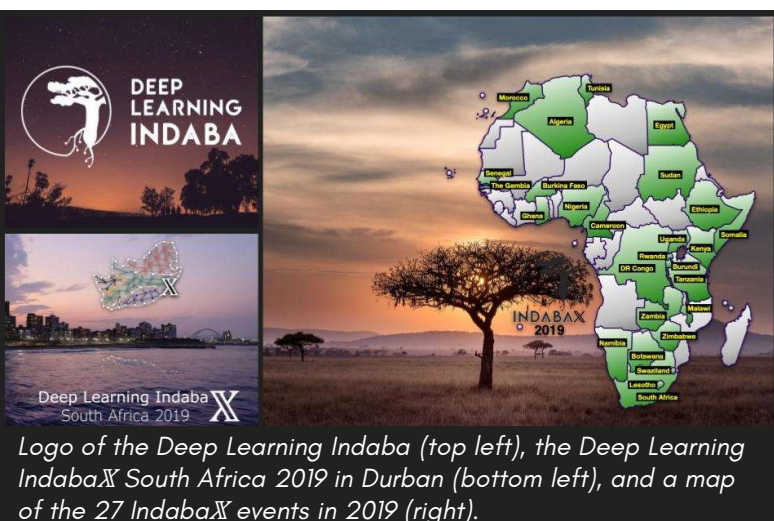


Durban beach front, the location of the South African Deep Learning Indaba^X

The world continues to change at a rapid pace, with **artificial intelligence** and machine learning increasingly playing an integral role in human advancements. These technologies are expected to shift power dynamics and shape the future in unprecedented ways in what is termed the "**4th industrial revolution**". Africa, in turn, is primed to leverage an ever-growing human capacity in order to make important contributions in the **machine learning community**.

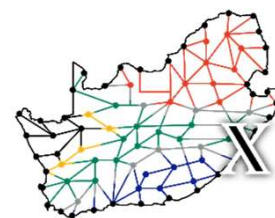
Born from the desire to diversify this community defining the future, the pan-African **Deep Learning Indaba** was founded in 2017 to spread knowledge and strengthen African capacity in machine learning. Seeded from the root conference, a diaspora of 27 **Deep Learning Indaba^X** events are spread across the African continent in 2019, each a local community-driven meeting disseminating knowledge wider and deeper.

Having been established in South Africa in 2017 and 2018, the Deep Learning Indaba will be held this year in Nairobi, Kenya. This presented the South African Indaba^X with the responsibility to fill the large gap left behind and become the pinnacle education and research-focused artificial intelligence conference in the country.

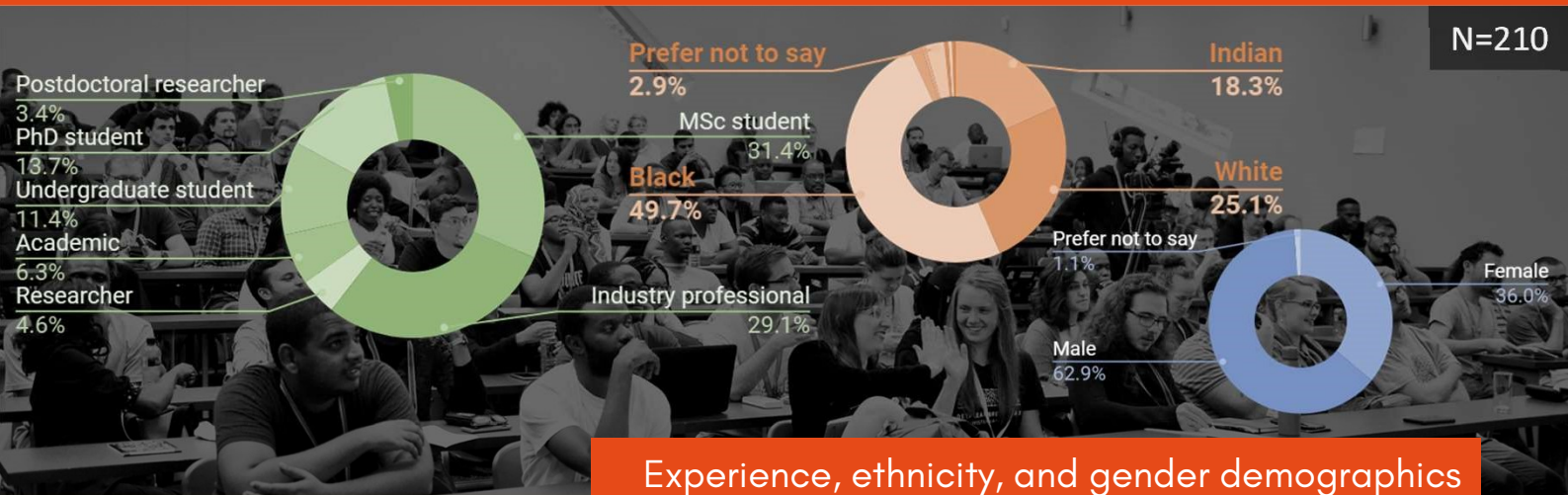


In building capacity for machine learning in South Africa, we sought to create lasting connections, drive innovative developments, and proactively influence how recent advances shape our society. The Deep Learning Indaba^X South Africa therefore presents an opportunity for students, researchers, and industry to come together from around the country to learn, share, and grow machine learning in South Africa and beyond.

DEEP LEARNING INDABA^X SOUTH AFRICA 2019



INTRODUCTION



With nearly 500 applications and a venue that seated far less, picking the top 210 candidates was tough. We **proactively targeted under-represented groups** from around the country, both for **delegates and speakers**. We also weighted applicants by their motivation to **gain knowledge** and their desire to **build connections**, which are fundamental factors in growing machine learning capacity in Africa. Ultimately, we picked people **passionate** about machine learning, giving a chance for those looking to enter the space as well as those looking to build expertise.

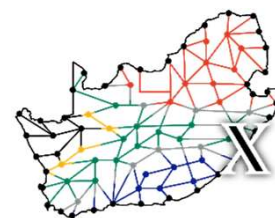
Over 74 hours*, the **210 delegates** laughed, learnt, shared, and planted roots to grow machine learning in Africa. With **73 travel fellowships** that fully funded students from around the country, we were able to service learners who may not have had the means to attend, exemplifying **inclusion** and **diversity**. This melting pot of participants embraced the opportunity to engage with top minds and future talents.

While we were limited to 210 attendees due to space constraints, we strongly believed this material must be made **freely available for sharing** with the wider machine learning communities scattered across the country. To achieve this aim, we had all the talks filmed (the speakers kindly agreed to share their slides and voices), a photographer to capture **serendipitous moments** (featured in these pages), and a stream of the plenary sessions for real-time virtual engagement.

Our goal of **inclusion** and **diversity** informed our aims to 1) **provide a practical introduction to machine learning for beginners**, 2) **create an exchange for sharing scientific knowledge amongst researchers**, and 3) **showcase cutting-edge applications, cognisant of ethical considerations, to industry**.

In this report, we look back at the event to share great moments and success stories, and to reflect on where there is room for improvement. Importantly, we built a strong foundation for continuation of the South AfricaIndaba^X so that inspired individuals can take the torch and make it shine brighter each year.

DEEP LEARNING INDABA^X SOUTH AFRICA 2019



PROCEEDINGS



AFRICA in AI setting the tone for the Indaba^X

Sunday the 14th of April saw South Africa's Deep Learning Indaba^X open for **registration**. Due to the large number of attendees and the packed schedule, this opportunity dedicated some time for mingling before bussing everyone to the opening event nearby!

The **opening event** had a DJ, dancing, and drinks plus food, friends, and new faces. Many found the start "awesome", "[a] great way to interact", "lively", but also "a bit loud". Entering a busy 3 days ahead, we wisely capped the official event at 10pm.

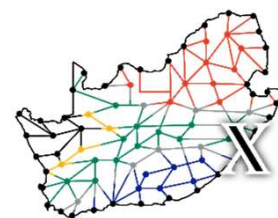
Early on Monday, we kicked off with a short **welcoming address** by Maria Schuld, Amira Abbas, and Prof. Francesco Petruccione who was speaking as the host. This lead to the first plenary session, a **panel** discussion on "**Africa in AI**" moderated by Christopher Currin and featuring a range of panellists with incredible diversity in so many meanings of the word (pictured above).



Clinton Reddy of the "North Beach" sponsor STANLIB asking a question to the panel.

While unconventional, having the panel at the beginning of the proceedings set the tone on a number of aspects for the event. First, it showcased the **diverse backgrounds** of everyone at the event: from academia and industry, Cape Town and Johannesburg, Vice Chancellor and MSc candidate, and many more. Second, it served to **define a narrative** as focusing on AI and its complexities from our African perspectives rather than Africa from an AI perspective. Third, with disagreements arising from the discussion, it showed the audience that even top people in the field **may not have all the answers** and can disagree amicably. Finally, it **fostered discussions throughout** the conference by making conversations and controversy welcomed from the very start.

DEEP LEARNING INDABAX SOUTH AFRICA 2019



PROCEEDINGS



Left, Tshilidzi Marwala presenting with the audience
Right, Andrew Saxe explaining a technical concept

Directly following the panel was our first **keynote** speaker Prof Tshilidzi Marwala, Vice Chancellor of the University of Johannesburg talking on the rationality of artificial intelligence machines.

After a short tea break, the proceedings split into three tracks: **Foundations**; **Research**; and **Industry, Policy, and Ethics**.



Edgar Jembere

Foundations was aimed at attendees early in their machine learning career or for those looking to learn a new technique such as *unsupervised learning* or *reinforcement learning*. These sessions were targeted as practical introductions where delegates could follow along on a computer, yet near the end demonstrate some advanced knowledge and powerful applications to entice further engagement beyond the session.

Research showcased our homegrown research activities across South Africa, featuring representatives of some major groups active in the machine learning sphere. Because of the variety of topics and attendee backgrounds, these talks typically included ample background knowledge before delving into results.



Maria Schuld

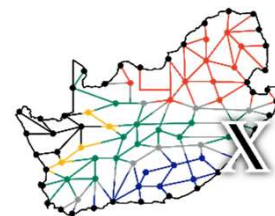


Pelonomi Moilola

Industry, Policy, and Ethics exhibited influential minds from diverse companies. The development, application, and impact of AI was poignantly discussed and engaged with by audience members both established and starting out.

The end of the first day saw the spotlight shift from invited speakers to attendees' research with a **poster session**.

DEEP LEARNING INDABAX SOUTH AFRICA 2019

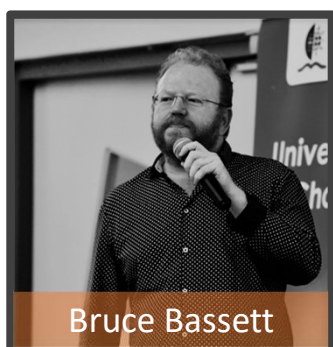


PROCEEDINGS



30 posters strewn across a corridor for lively conversations and probing insights

Tuesday morning broke with a **keynote** by Andrew Saxe, a post-doctoral research associate in Deep Learning, Neuroscience, and Psychology at the University of Oxford. The in-depth insight he gave into the interplay between deep learning and generalisation was closely followed by the audience to form a strong start to the second day of proceedings.



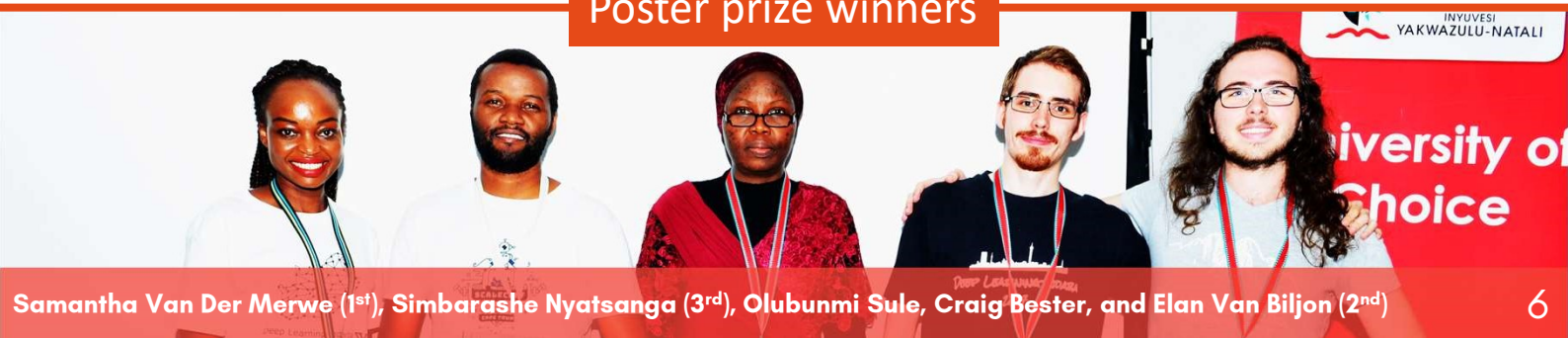
Bruce Bassett

Again the day split into the three parallel tracks until the final talk and **keynote** by Bruce Bassett, who is a professor of Mathematics at the University of Cape Town/AIMS/SAAO, Head of AIMS cosmology and machine learning group, and Head of Data Science and Machine Learning at SKA South Africa. His fun and adventurous 4-part talk outlining AI developments and its future was welcomed as a fitting ending to the formal presentations.

Immediately afterwards, the poster prizes were awarded: 1st place **Samantha received a full travel fellowship to the Deep Learning Indaba** in Nairobi, Kenya to present her work on *Predicting social unrest events in South Africa* to 1000+ participants! 2nd place Elan won a full pass to the AI Expo Africa in September (valued at R4500) courtesy of Cortex Logic and 3rd place Simbarashe won the "Deep Learning" textbook courtesy of the organizing committee member Amira Abbas.

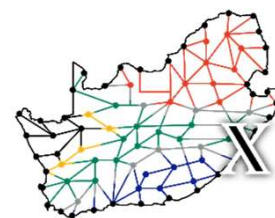
But while the day had ended at 17:30, the hackathon was about to begin!

Poster prize winners



Samantha Van Der Merwe (1st), Simbarashe Nyatsanga (3rd), Olubunmi Sule, Craig Bester, and Elan Van Biljon (2nd)

DEEP LEARNING INDABA $\times$ SOUTH AFRICA 2019



HACKATHON + UNCONFERENCE



After two days loaded with learning, the participants were eager to get their models dirty with data during the **HACKATHON**. Working with committed sponsors, we could offer the participants a range of problems for them to tackle over 18 hours.

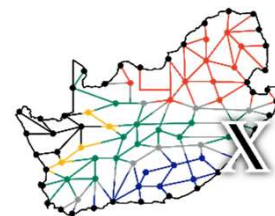
Open Data Durban provided the Indaba $\times$ hackathon team with the South African Census data from 2011 for a beginner challenge we crafted focused on basic concepts such as data pre-processing, regression, and classification. **Isazi Consulting** challenged more experienced participants to develop a model that used image segmentation and classification to convert images of numbered sequences into their text equivalents. **IBM** provided a dataset comprising of DNA sequences and tasked the participants to use deep learning to predict whether the DNA sequence was from a zoonotic or non-zoonotic virus*. Finally, the **Aerobotics** team released a dataset of orchard imagery with the goal of training a Cycle GAN to segment the pixels in images between a set of classes. Sponsor-provided tutors were around to help too!

With the incentive of some pizza (thanks **Data Prophet**), snacks (thanks James Allingham), some high performance computers (thanks **CHPC**[#]), interesting challenges, and amazing prizes up for grabs, the participants worked late into the night at the inaugural Indaba $\times$ South Africa hackathon.

The third day began with hackers returning to the UKZN LAN venue as well as many others eager to engage on interesting topics not covered by the schedule during the **UNCONFERENCE**. Groups formed and converged around diverse topics such as “causality”, “GANs”, “ML and the scientific method”, and more which sprouted throughout the allotted time. This provided a semi-structured form of knowledge transfer that is less formal than a talk, yet with experts naturally leading discussions.

At lunch time, the unconference and hackathon came to a close and in the final event of the conference, we began the hackathon prize giving.

DEEP LEARNING INDABAX SOUTH AFRICA 2019



HACKATHON



Among the exciting prizes was a pair of graphics cards for deep learning



For the Open Data Durban challenge, 1st place, **Lerato Mnguni**, won the "Deep Learning" textbook by Ian Goodfellow, Yoshua Bengio and Aaron Courville courtesy of Data Prophet and 2nd Place, **Tarirai Chani**, won a R1000 Exclusive Book's voucher thanks to the UZKN Big Data Flagship.



Aerobotics awarded 1st and 2nd place to two teams: **Computer-Vision (Ronald and Omran)** and **No Name Team (Katlego, Ditebogo, Moyahabo, and Crestinah)**, who each took home an array of amazing textbooks: *The Elements of Statistical Learning*, *Pattern Recognition with Machine Learning*, *Machine Learning: A Probabilistic Perspective* and *Reinforcement Learning: An Introduction*.

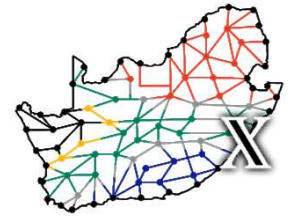
For the Isazi Consulting challenge, 1st place went to **Michael Gant** and **Guillaume Odendaal** who each won a GEFORCE RTX 2060 GPU and 2nd Place, **Nangue Tasse Geraud**, won R1000 in cash.



For the IBM challenge, **Mkhupuli Ncube** grabbed 1st place with a cash prize of \$250, Aerobotics's own **Scott Glover** came 2nd with \$150 and finally, **Craig James Bester** and **Christopher Dunderdale**, placed 3rd with a prize of \$100. Special thanks to the Eck Institute for sponsoring the IBM prize.



DEEP LEARNING INDABA^X SOUTH AFRICA 2019



SPONSORS



Cortex Logic came ready with mugs and smiles

Without the support of our amazing and generous sponsors, the Deep Learning Indaba^X South Africa 2019 would not have been able to reach its aims dreams.

TIERS

Deep Learning Indaba



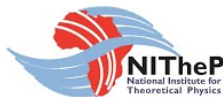
Umhlanga Rocks



North Beach



South Beach



New Pier



Hackathon

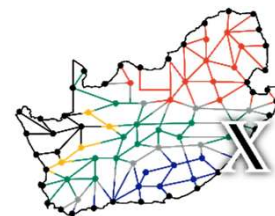


Community



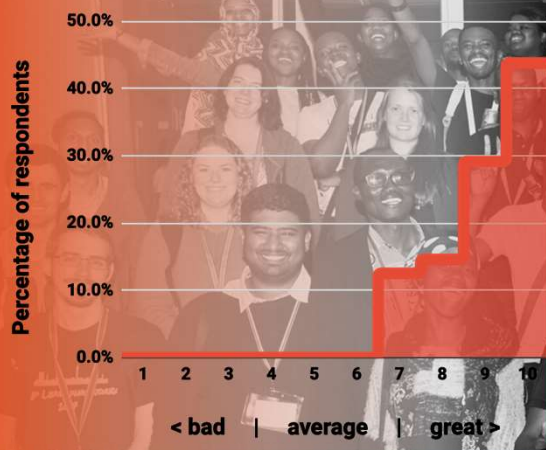
Machine Intelligence Institute of Africa
Transform Africa through Machine Intelligence

DEEP LEARNING INDABA $\times$ SOUTH AFRICA 2019



FEEDBACK & IMPROVEMENTS

How would you rate the Indaba $\times$?



73 travel fellowships awardees showing their love and appreciation.

After an exhausting event, it was time to rest, digest, and reflect.

Nearly 75% of attendees rated the Indaba $\times$ as a 9 or 10 (out of 10) **overall**, with many finding the content "very relevant" and the speakers "knowledgeable" with "some great insights." Attendees had some powerful key moments from the event that they shared including that South Africa has a thriving and diverse machine learning landscape ,

Tshilidzi's keynote, the panel as well as the research talks. ,

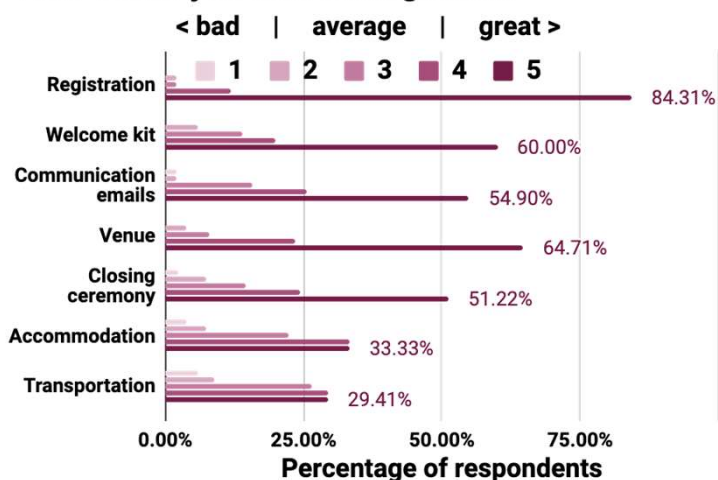
networking ... really motivated me...

as well as some inspiration that these events do make a difference:

hands down the best experience of my life!

I will forever be grateful for the opportunity to learn and grow .

How would you rate the logistics?

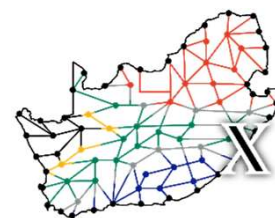


While much of the **logistics** seemed well-rated, the *registration process* in particular, some thought the *communication* could have been better (e.g. sending the schedule earlier), the *venue* needed more power outlets, and the *accommodation* and *transportation* needed greater attention for those who received travel fellowships. While we were fortunate to secure sponsorship to fund 73 students from around the country to attend, with special acknowledgement to the DST-NRF CoE-MaSS* grant, we could have done better.

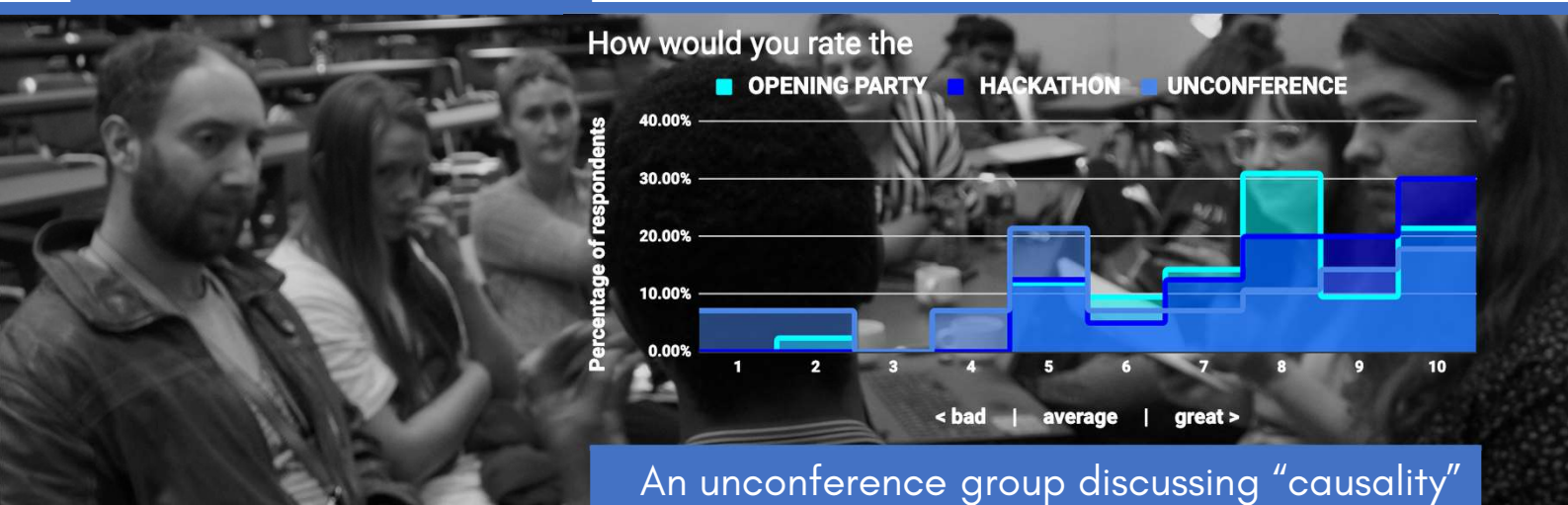


Helpers making the registration flow smoothly

DEEP LEARNING INDABA $\times$ SOUTH AFRICA 2019



FEEDBACK & IMPROVEMENTS



The **opening party** was well-received with many enjoying the food, atmosphere, and spontaneous conversations. The main criticisms were that it was too loud for some and that with no ice-breakers, some cliques formed. While hard to overcome societal tendencies and create fun ice-breakers, we are sure to improve for next year.

In 2018, the Indaba $\times$ Western Cape team had investigated organising a **hackathon**, but it was not to be. One of the key lessons was how long it takes to get data approved (even if it is publicly available). The Indaba $\times$ South Africa team therefore started the process at the end of 2018 (before official Indaba $\times$ announcements) by talking to companies with which to partner. This worked well as 70% thought the hackathon was great (8 or higher). Some "hackers" did think all the problem were too advanced, yet we did try to include educational aspects in some of the problems. In future, it would be good to better align the hackathon sponsors in terms of prizes and starter-code expectations.



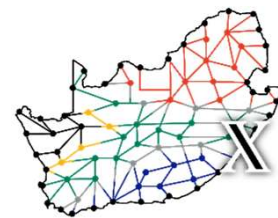
Hackathon "hackers"



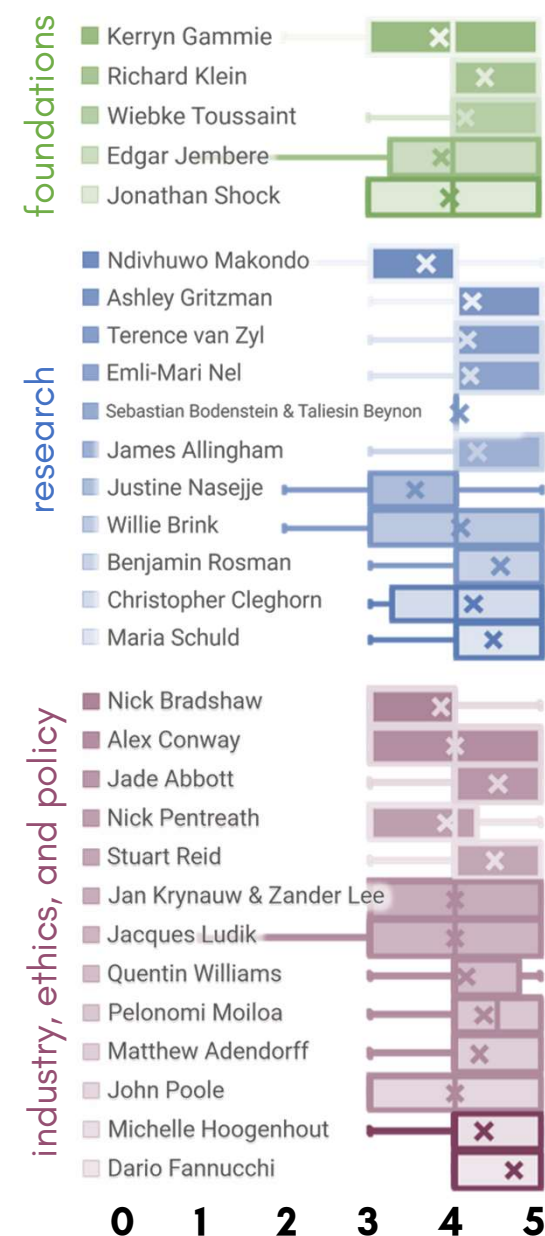
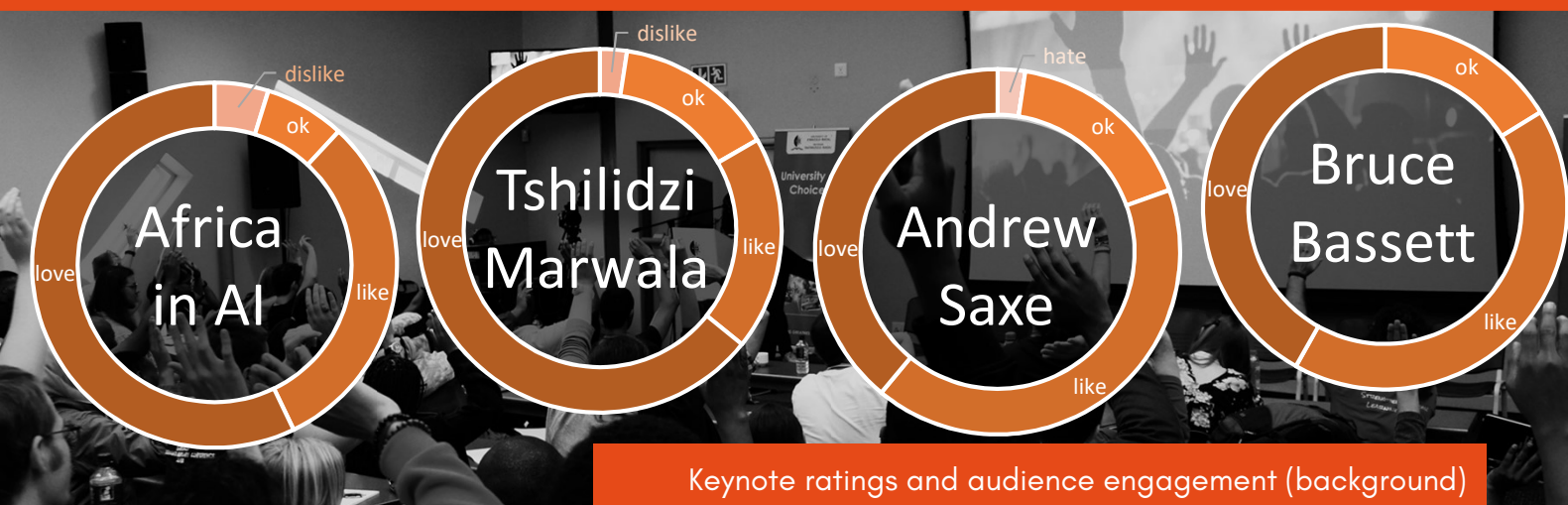
Unconference group

The **unconference** was a novel addition to Deep Learning Indaba $\times$ schedule as it is a relatively new concept of having the audience drive the agenda and structure the content. While some participants found the unconference discussion on causality a highlight, others didn't find a critical mass in their own topic of interest. We had asked the attendees to provide suggested topics for the unconference beforehand and then decided final groups of discussion on the day. With a diverse array of topics which came up, it was difficult to narrow down the suggestions. Advice for future unconferences, which we think should be included if given the time, is have more time at the beginning to distil topics of interest, create a more structured agenda where people can move between topics at intervals, and have a few more leaders that can guide the event.

DEEP LEARNING INDABA^X SOUTH AFRICA 2019



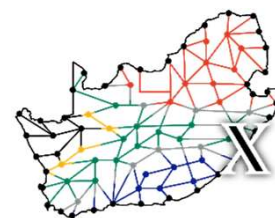
FEEDBACK & IMPROVEMENTS



The **talks** were very well received by the delegates with particularly high scores achieved by the industry track. All tracks were roughly equally popular with the attendees, and the keynotes were met with **overwhelmingly positive feedback**.

Not surprisingly, the comments reflected the large diversity of the audience, and some missed the technical depth in **high-level talks**, while others would have liked more introduction to the **technical presentations**. Andrew Saxe, for example, elegantly navigated through a wealth of technical and highly relevant results, and so was evaluated as "mind-blowing", while some delegates wished for "more explanations". Another example is the panel discussion, that many found "eye-opening" but also some criticised it for being too "high-level".

An important point of internal criticism, coming from the organising team, is the **diversity of the speakers**. Although we put a lot of effort into racial and gender balance, with the result being much ahead of any international machine learning conference, we hope to feature an even more diverse line-up in the future. A way to encourage this would be to allow more MSc/PhD students to present. Furthermore, we have started an initiative to have a shared pool of speakers with the other 26 Indaba^Xs to help achieve a broader representation.



FEEDBACK & IMPROVEMENTS



Ashley Gritzman guiding gazes and pointing out mistakes (in CNNs)

While overall a success, we learnt some important **behind-the-scenes lessons** along the way including the difficulty in handling finances partially based on university cost centres, organising ground transportation for sponsored students, and ensuring the commitment of sponsors and the host institution. To improve next year's organisation handling of finances, we have set up a non-profit to facilitate more rapid payments.

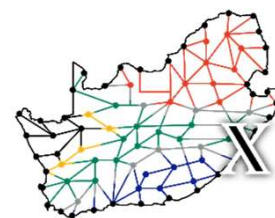
Balancing the different levels of technical backgrounds is a major challenge for a conference that understands itself both as a platform for **excellent academic exchange**, but also as a **catalyst to form a community**. We feel the focus on accessibility of content was well-suited to this type of event, but some clearly want to see a more technical-focused research conference, in the vain of the big international conferences, in the near future. We believe that the South African Indaba $\times$ should aim to steadily increase the technical content and level of the talks; to grow the content with the growth of the academic landscape in South Africa.

The Indaba $\times$ initiative is a means to experiment with **new ways of building capacity** in machine learning and so the focus tends to be broad to grow the **many groups** which make up our **community**. Given that there are few **research-focused machine learning conferences** in South Africa, and Africa, it would incredibly powerful to host one harnessing the spirit of an inclusive community – *the Ubuntu* proceedings of Artificial Intelligence* (UAI or “You n I”). One idea would be to reach out to IEEE South Africa with a view for submitted posters, with extended abstracts, to be peer-reviewed and published so that there is further incentive to attend as well as to attract more international attendees.

The proceedings could stay connected with the Indaba $\times$ education roots with a short and intense summer school beforehand with fewer attendees but higher engagement, akin to the Machine Learning Summer School in Stellenbosch. The attendees can then attend the proceedings for free and gain exposure to a conference setting.

* Ubuntu is a nebulous concept of common humanity and oneness often summarised as “I am, because you are”

DEEP LEARNING INDABA^{XX} SOUTH AFRICA 2019



ORGANISATION



Celebrating the end of a successful Indaba^{XX}!

A group of **highly motivated team** players from across the country rose to the occasion of organising **one of the biggest Indaba^{XX}** events on the continent and one of the biggest artificial intelligence and machine learning conferences in the country. Passion, planning, and patience meant the team worked together to pull off a 3-day event that **inspired** and **motivated** attendees with few hiccups.

While spread across the country, and with some having never met IRL*, the modern age has meant remote working is true for conference organisation too. It helped using a central repository of information, having leaders clearly documenting and communicating decisions on a **Discord[#] server**, meeting on a weekly **video call** for group rapport and discussing larger issues, and creating a **message group** for "firefighting" during the event. Starting the **planning early** was paramount to the event's success, especially with regards to **organising a hackathon**. Local organisers, of the host institution, were necessary to acquire the venue and set the dates during holidays to facilitate sponsored accommodation in nearby residences. *Thank you Prof Francesco Petruccione for being willing to host and Dr Maria Schuld for leading the team as a local organiser.*

LIST OF ORGANISERS

AMIRA ABBAS

University of KwaZulu-Natal

CHRISTOPHER CURRIN

University of Cape Town

FRANCESCO PETRUCCIONE

University of KwaZulu-Natal

JAMES ALLINGHAM

Wolfram Research

STEVEN JAMES

University of the Witwatersrand

ANBAN PILLAY

University of KwaZulu-Natal

ANNIKA BRUNDYN

University of Cape Town

SASHA NAIDOO

Nedbank

MARIA SCHULD

University of KwaZulu-Natal

ALEX CONWAY

NumberBoost

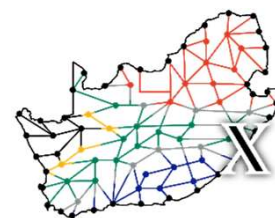
ANDREW PASKARAMOORTHY

University of the Witwatersrand

SICELUKWANDA ZWANE

University of the Witwatersrand

DEEP LEARNING INDABA^X SOUTH AFRICA 2019



BUDGET



An audience engaged between talks

The sponsorships were kicked off with the DST-NRF CoE-MaSS, thanks to some Deep Learning Indaba organisers, which meant we knew we could scale this Indaba^X to include travel fellowships, a poster session, a hackathon, and an opening party. While registration fees can offset expenses, it does require additional admin and more uncertainty. Fortunately, we managed to balanced the books almost perfectly.

EXPENSES	
Item	Cost
Travel Fellowships	-R225,056.24
Opening Party	-R86,900.00
Food	-R84,086.00
Venue & AV	-R71,000.00
Swag	-R38,992.46
Video Recording	-R23,000.00
Speakers + Organisers travel	-R21,287.08
Hackathon	-R4,000.00
Poster session	-R2,639.00
Admin	-R500.00
TOTAL	-R557,460.78

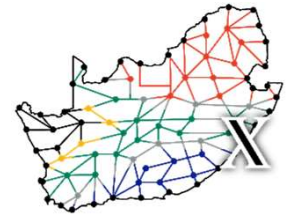
REGISTRATION FEES	
Source	Amount
Industry Registration Fees	R70,000.00
Academic Registration Fees	R16,250.00
TOTAL	R86,250.00

SPONSORSHIPS (excl. prizes)	
Partner	Amount
DST-NRF CoE-MaSS grant	R100,000.00
UKZN South African Research Chair in Quantum Information Processing and Communication	R71,000.00
STANLIB	R50,000.00
Cortex Logic	R50,000.00
alis_	R50,000.00
NITheP - National Institute for Theoretical Physics	R35,000.00
Deep Learning Indaba	R25,000.00
IBM Research	R15,000.00
Capitec Bank	R15,000.00
Interactive Solutions	R15,000.00
KRISP - KwaZulu-Natal Research Innovation and Sequencing Platform	R10,000.00
Psybergate	R10,000.00
Data Prophet	R10,000.00
Numberboost	R5,000.00
ConversionScience	R5,000.00
Xineoh	R5,000.00
TOTAL	R471,000.00

TOTAL

-R210.78

DEEP LEARNING INDABA $\times$ SOUTH AFRICA 2019



CONCLUSIONS



The Deep Learning Indaba $\times$ South Africa

With the **rapid advancement of AI** across the world, it is imperative that Africa not only has a voice, but an authority in the field. One way to stand out is by embracing “African problems” through our domain knowledge and unique contexts to **solve local problems that impact global solutions**. Doing it alone is never an option. The (South) **African AI community** is growing at an exponential rate year-on-year. The connections formed at these events form a network of people integral to growing the field and each other.

Collaborations between local and international universities and companies are vital to growth and development. In addition, the noticeable number of international African students at the Deep Learning Indaba $\times$ shows hunger for **greater partnerships** on the continent. Still, inter-country policies need to be addressed to facilitate better communication between collaborators. Fortunately, the Deep Learning Indaba movement is a means to **strengthen African machine learning** through knowledge transfer and connection building.

These events last beyond the schedule and reach beyond the attendees. The passion of everyone involved creates power. Powerful algorithms. Powerful connections. Powerful people that grow the space and **seed the next event**.

We truly appreciate everyone involved in creating a special event. From those **attending defining the experience** to the **sponsors supporting** the organisers in executing their vision. We are thrilled to see so many people eager to push what is possible and create a brighter future.

From this experience, which we poured our souls into, we have celebrated our successes and learnt from our failures, shared some incredible laughs and formed new or stronger bonds. Now it is time we look forward and see how you can play your own role in making **Deep Learning Indaba $\times$ South Africa 2020** that much more special.